The Effect of Difficulty Labels on Puzzle Engagement

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0.1 Introduction

How difficult a task feels can shape how much effort a person puts into it. This experiment asked whether showing participants a difficulty label before a logic puzzle — telling them it is easy or hard — changes how long they stay engaged with it. Participants were given the same set of hard puzzles and randomly assigned to one of three conditions: no label, a hard label (accurate), or an easy label (understated). We compared mean time engaged across the three groups to test whether the label changed behavior. Prior work on perceived task difficulty suggests that the information people have about a task, along with their expectations going in, influences how much effort they invest. By varying the label shown before each puzzle, we manipulated those expectations and observed the effect on engagement.

0.1.1 Theory

Priming is when exposure to one stimulus influences how you respond to the next. Research on semantic priming has shown that people recognize related words faster when preceded by an associated cue (Meyer and Schvaneveldt 1971), suggesting that prior context shapes how we process what comes next.

Kahneman (2011) extends this beyond words: he argues that System 1 — the fast, intuitive mode of thinking — is continuously shaped by whatever context is available at the moment of judgment. A difficulty label shown before a task is exactly this kind of cue. Before a participant even reads the first clue, System 1 has already formed an expectation about how hard the task will be and how much effort it deserves. By the time deliberate reasoning engages, the frame is already set.

When that frame is wrong — when someone primed with "easy" encounters a genuinely hard puzzle — the mismatch creates cognitive friction. Baumeister et al. (1998) showed that sustained cognitive effort depletes a limited self-regulatory resource, lowering persistence over time. A violated expectation compounds this: participants are not just working against the puzzle, they are also working against their own prior expectation of how this should feel. That additional load, we argue, lowers the threshold for giving up.

0.1.2 Hypothesis

We hypothesize that the two label conditions affect time engaged in opposite directions relative to the no label control. Participants shown the hard label receive accurate information about task difficulty and may mentally prepare for sustained effort, increasing persistence. Participants shown the easy label encounter a mismatch between their expectation and their experience: the puzzle is harder than primed, which we expect to produce frustration that shortens engagement. When participants expect a task to be easy and immediately find it is not, the experience may feel like failure rather than challenge, lowering the threshold for giving up and skipping. Formally:

$$H_1 : \mu_{\text{hard}} > \mu_{\text{no label}}$$

$$H_2 : \mu_{\text{easy}} < \mu_{\text{no label}}$$

$$H_3 : \mu_{\text{hard}} > \mu_{\text{easy}}$$

where μ denotes mean time engaged per puzzle round.

0.2 Power Calculation

To determine the necessary sample size, we conducted a simulation-based power analysis rather than relying on a closed-form analytic formula. This approach allows us to model the multi-round panel structure of the experiment and account for participant-level clustering.

We set a control baseline average solve time of $\mu = 270$ seconds. Variance was decomposed into a between-person component ($\sigma_u = 60$ seconds) and a within-person residual ($\sigma_e = 67$ seconds), yielding a total standard deviation of $\sigma = \sqrt{\sigma_u^2 + \sigma_e^2} \approx 90$ seconds, chosen conservatively to capture natural individual variation in performance. We specified treatment effects relative to the no label control in opposite directions for the two label arms: a $+25$ second increase for the hard label arm, reflecting anticipated persistence when participants are accurately primed for difficulty, and a -25 second decrease for the easy label arm, reflecting earlier disengagement when participants encounter a task harder than the label suggested.

We simulated 500 iterations at escalating sample sizes. For each iteration, a mixed-effects model was fit with puzzle fixed effects and a participant-level random intercept:

$$Y_{it} = \beta_0 + \boldsymbol{\beta}_{\text{trt}}\text{Treatment}_i + \boldsymbol{\gamma}_t + u_i + \varepsilon_{it}$$

where $\boldsymbol{\gamma}_t$ are puzzle fixed effects and $u_i \sim N(0, \sigma_u^2)$ is a participant random intercept. Because data are generated exactly from the model, standard mixed-model t-tests are used to assess significance; cluster-robust standard errors are applied in the actual analysis but are not needed for simulation where the variance structure is known.

At $\alpha = 0.05$, power to detect the hard label effect crosses the 80% threshold at 120 participants per arm (360 total), and the easy label effect crosses 80% at 110 per arm (330 total). The harder-to-detect effect requires 120 per arm (360 total).

The final analytic sample of 256 completers falls below the pre-specified target of 360. Exclusions for behavioral flags and attrition reduced the sample beyond what was anticipated at the design stage, and arm sizes are unequal (No Label: 87, Easy Label: 80, Hard Label: 89), which modestly reduces power relative to equal allocation. Beyond sample size, the analytic sample pools two recruitment populations that differ systematically in engagement conditions: Prolific participants completed the experiment in a structured, incentivized setting with low distraction, while organic participants, recruited through personal networks, varied more in motivation, setting, and distraction level. This heterogeneity inflates within-group variance beyond what a single σ captures, further eroding effective power. Both the sample shortfall and the variance inflation from pooling two behaviorally distinct populations should be kept in mind when interpreting null or marginal findings.

0.3 Study Design

0.3.1 Procedure

The experiment was delivered as a browser-based web application at anotherpuzzle.io. The application was a single-page JavaScript game engine that managed the complete participant session from recruitment intake through behavioral data collection without any researcher intervention at runtime. All data were logged directly to a cloud database (Firestore) at both the session and round levels.

Upon arriving at the application, participants were presented with a consent screen. Participants who agreed were immediately and invisibly assigned to one of three treatment arms before proceeding. Following consent, participants completed an eight-item survey collecting baseline covariates: age, gender, education level, quantitative exposure, current energy level, puzzle frequency, self-rated puzzle skill, and puzzle enjoyment. After the survey, participants were shown a brief instruction screen and then clicked "Start the Challenge," which triggered randomization of puzzle order and the creation of the participant's session record.

The game engine logged behavioral data at the round level. For each round, it recorded: total time engaged (a continuous timer that paused automatically during tab switches and participant-initiated breaks), number of answer attempts, whether the first attempt was correct, time to first attempt, tab-switch count and total time paused, rage-click events, and whether the round ended in a solve or a skip. Skip was enabled after 30 seconds of engagement, preventing immediate abandonment without any attempt. At the session level, the engine recorded device type, puzzle presentation order, recruitment source, and session completion status.

0.3.2 Treatment

The experimental treatment was a difficulty label presented to participants before each logic puzzle. The hard label was expected to prompt mental preparation for a demanding task, increasing persistence. The easy label was expected to create an expectation mismatch: participants anticipating an easy task would encounter greater difficulty than expected, producing frustration that shortens engagement before participants choose to skip.

Participants were randomly assigned at session entry to one of three conditions: (1) no label, (2) easy label, or (3) hard label. The assigned label was shown before every puzzle in the session, so a participant assigned to the hard label condition saw that label at the start of each of the five rounds. Within the application, all other aspects of the experience were held constant across arms. The interface, instructions, and the five logic puzzles were identical across conditions, so differences in behavior across treatment groups can be attributed to the difficulty label rather than any other feature of the experience.

Each session consisted of five logic puzzles presented in randomized order: a number sequence, an alphabet wheel, a shape sequence, a hybrid shape-number puzzle, and a tile sequence. All five were designed to be intentionally challenging, though they varied in how difficult participants found them in practice.

```
## Error in `select()`:
## ! unused arguments (Puzzle, no_label, easy_label, hard_label, overall_raw)
```

Skip and solve rates differed meaningfully across puzzle types, as shown in Table **??**. Puzzle order was randomized across participants to prevent any one puzzle from systematically falling at the same point in the session, and the analysis accounts for puzzle-level differences and within-participant correlation across rounds.

Treatment delivery was controlled entirely through the application, and the data contain no indication that any participant was shown a label other than their assigned condition; we therefore identified no treatment non-compliers. However, the application did not independently log the randomized assignment alongside the label rendered on screen for each round, so perfect delivery could not be independently verified. Future experiments should log these separately to enable a formal compliance check.

0.4 Participants

The experiment database contained 398 unique participants. Eight were excluded as returning or replay players whose devices had previously completed a full session, leaving 390 first-time participants. An additional

27 had no retained gamelog records and were excluded, yielding 363 participants who started at least one puzzle round.

Twenty-one participants were excluded for eligibility: 8 were under 18 and 13 were rejected through Prolific's quality review. Of the remaining 342 eligible participants, 39 were removed for behavioral or data-quality reasons:

Behavioral exclusions

1. Total tab-switch pause time exceeded 600 seconds across the session, indicating the participant was away from the browser for an extended period.
2. At least one puzzle showed an idle walk-away pattern: the participant skipped without switching tabs and with time engaged above the 97th percentile of the skip-time distribution for that puzzle, suggesting they left the page and returned.
3. All five puzzles were skipped within 30 to 35 seconds with no answer attempts, indicating the participant waited out the skip-unlock timer on every round rather than engaging with the puzzles.

AI assistance exclusions

4. Self-reported use of AI assistance in the post-session honesty check.
5. Hard puzzles solved on the first attempt faster than population benchmarks for that puzzle (any single puzzle below the 5th percentile, or two or more below the 10th), subject to manual review. Flagged participants were retained if their fast solve times were consistent with their relative performance across all rounds, as genuine skill tends to produce proportionally fast times across puzzles rather than selectively fast times on isolated ones.

Attrition, defined as beginning treatment but failing to complete all five rounds, accounted for 47 participants. The final analytic sample is 256 completers across three arms: No Label (n = 87), Easy Label (n = 80), and Hard Label (n = 89). Figure 1 shows the full participant flow.

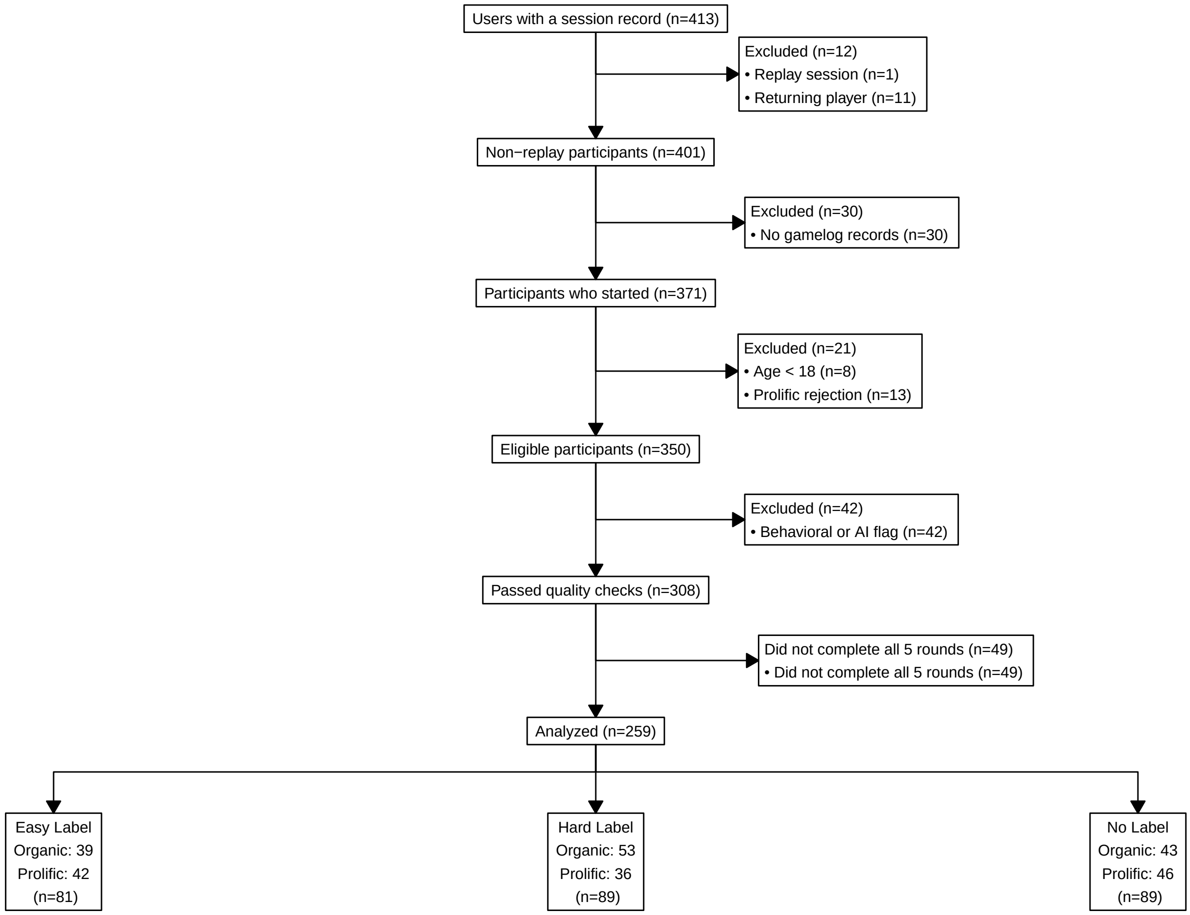

Figure 1: Participant flow through the experiment.

0.5 Potential Outcomes

Each participant has three potential outcomes — how long they would spend on the puzzles under each label condition: Y_i(no label), Y_i(easy label), and Y_i(hard label). We only ever observe one of these per person. Because participants were randomly assigned, the treatment someone received is independent of what their outcome would have been under a different condition, which means we can estimate average outcomes for each arm using the observed group means.

The main comparisons we care about are the treatment effects of each label relative to the no-label control, and the direct comparison between the two label arms. Each τ below is the difference in group means (μ) between two arms:

$$\tau_{\text{easy}} = \mu_{\text{easy}} - \mu_{\text{no label}} = E[Y_i(\text{easy label})] - E[Y_i(\text{no label})]$$

$$\tau_{\text{hard}} = \mu_{\text{hard}} - \mu_{\text{no label}} = E[Y_i(\text{hard label})] - E[Y_i(\text{no label})]$$

$$\tau_{\text{hard vs. easy}} = \mu_{\text{hard}} - \mu_{\text{easy}} = E[Y_i(\text{hard label})] - E[Y_i(\text{easy label})]$$

The no-label arm is the reference group — participants who saw no framing at all. The first two comparisons ask whether giving someone any difficulty expectation changes their engagement relative to having none. The third asks whether the direction of the label matters: we expect the easy and hard labels to push behavior in opposite directions, so $\tau_{\text{easy}} < 0$ (frustration leads to less engagement), $\tau_{\text{hard}} > 0$ (preparation leads to more), and $\tau_{\text{hard vs. easy}} > 0$.

0.6 Randomization and Attrition

0.6.1 Randomization

Treatment was assigned entirely client-side at the moment a participant clicked "I understand – let's go" on the consent screen. The game engine drew a single uniform random variate and mapped it to one of three arms with nearly equal probability:

```
r = Math.random()
Treatment = r < 0.333 ? 'no_label'
          : r < 0.666 ? 'hard_label'
          : 'easy_label'
```

`Math.random()` draws a number between 0 and 1 uniformly, with the unit interval cut into three equal slices. Each arm was assigned with approximately one-third probability. The treatment draw was written to a `localStorage` draft immediately after assignment. Participants who closed and returned mid-session were shown a "Welcome back" screen that restored their original treatment from the draft, ensuring the draw was not re-rolled on page reload. A participant's IP address was additionally captured and stored under a separate `localStorage` key to detect whether a device had previously completed a full session. Returning devices were routed to a shared-device screening question where participants self-reported whether they were a new or returning player. Together, the IP-based flag and the self-report allowed us to distinguish true first-time sessions from replays and from new participants sharing a household device, retaining only genuine first sessions.

0.6.2 Recruitment Populations

We recruited participants from two different sources, and we track this throughout the analysis because recruitment context likely affects both dropout and how people engage with the task. Organic participants were recruited informally through personal networks — friends, family, and word of mouth — with no financial incentive to complete the study. Prolific participants were recruited through the Prolific platform, which pays participants for completing studies. Prolific workers tend to be experienced study participants who are used to following task instructions carefully and completing what they start.

0.6.3 Attrition

Attrition is defined as beginning treatment but failing to complete all five rounds. A participant is counted as an attritor if they have a user record in the database, created at the moment they click "Start the Challenge" and are shown a Round 1 label card, but their session was never marked complete. The variable `sessionComplete` is set to TRUE only when a participant reaches the results screen after resolving all five rounds, each ending in either a correct answer or a skip. Participants who passed through the survey and opted out before clicking "Start the Challenge" are not counted as attritors because no user record is created for them and they were never exposed to the treatment.

0.6.3.1 Differential Attrition in Organic and Prolific Subsamples The attrition analysis fits logistic regression models predicting dropout on treatment arm, estimated separately for the full sample and the organic subsample, with no label as the reference arm. The Prolific subsample had zero dropout across all arms and is reported descriptively only.

```
## Error in `select()`:
## ! unused argument (-source)
```

Among organic participants, dropout rates were 30.6% in the no label arm, 29.1% in the easy label arm, and 20.9% in the hard label arm. The easy label showed no difference from the control (OR = 0.928, p = 0.855). The hard label arm had a suggestive but non-significant reduction in dropout relative to no label (OR = 0.598, p = 0.207); notably, the primary model found the hard label went in the opposite direction from what we expected on engagement time. Dropout rates did not differ significantly across arms, so we did not find evidence that differential attrition meaningfully biased the primary outcome estimates. If the pattern were real, the no label arm losing more participants would leave its completers as a more selected group, which would compress the estimated differences between arms. The Prolific subsample had zero dropout across all three arms, which was expected given that Prolific participants are paid to complete studies (see Table **??**).

0.7 Completer Analysis & Covariate Balance

The primary analysis estimates average treatment effects among the 256 participants who completed all five rounds. Restricting to completers introduces potential attrition bias if dropout is related to treatment. As noted above, organic dropout was suggestively lower in the hard label arm, though not statistically significant. If that pattern reflects a real effect, the no label arm would lose a disproportionately higher share of participants, leaving its completers as a more select group and compressing estimated treatment effect differences toward zero.

Table 1: Covariate Balance Across Treatment Arms

	Age	Quant Exp	Puzzle Skill	Puzzle Enjoy	Energy	Recruit Source	Desktop	Female
Easy Label	−1.659	0.131	−0.160	−0.022	−0.093	0.002	0.066	0.161*
	(1.803)	(0.187)	(0.153)	(0.164)	(0.154)	(0.077)	(0.076)	(0.076)
Hard Label	0.753	−0.157	−0.247	−0.202	−0.090	−0.112	0.045	0.090
	(1.760)	(0.183)	(0.149)	(0.160)	(0.150)	(0.075)	(0.074)	(0.075)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

OLS coefficients vs. no-label reference arm. SEs in parentheses.

Table 1 presents covariate balance across the three treatment arms among completers. Each column reports an OLS regression of one baseline characteristic on treatment indicators, with no label as the reference arm. None of the easy label or hard label coefficients reach conventional significance thresholds on any covariate: age, quantitative exposure, puzzle skill, puzzle enjoyment, energy level, recruitment source, device type, or gender. The R^2 values are near zero throughout, confirming that treatment assignment explains essentially none of the variance in any baseline variable. This balance supports the validity of the completer analysis: observed differences in outcomes across arms are unlikely to reflect pre-existing differences in participant composition.

0.8 Results

0.8.1 Behavioral Differences by Recruitment Source

To assess whether recruitment source introduced systematic differences in data quality or engagement patterns, we examined four behavioral indicators across Prolific and organic participants: across-person variability in log time engaged, within-person variability in log time engaged, tab switch frequency, and total time spent with the browser tab inactive.

Table 2: Behavioral Differences by Recruitment Source

	Dependent variable:			
	Across-Person SD	Within-Person SD	Tab Switches	Time Paused (s)
	(1)	(2)	(3)	(4)
Prolific (vs. Organic)	−0.009	-0.260^{***}	0.136	-24.145^{***}
	(0.039)	(0.037)	(0.589)	(8.563)
Constant	0.427^{***}	0.896^{***}	1.896^{***}	40.920^{***}
	(0.027)	(0.025)	(0.408)	(5.925)
Observations	259	259	259	259
R^2	0.0002	0.162	0.0002	0.030
Adjusted R^2	−0.004	0.159	−0.004	0.026
Note:				*p<0.1; **p<0.05; ***p<0.01

Across the four indicators, two differences emerge. Across-person variability in log time engaged does not differ significantly between groups, suggesting that the two subsamples are comparable in their overall engagement levels. Within-person variability, however, is significantly higher among organic participants: organic participants were more erratic across their five puzzle rounds than Prolific participants. One plausible explanation is that organic participants, recruited informally through personal networks, varied more in their environment, distraction level, and motivation from one puzzle to the next. A participant might have been fully focused on one round and interrupted on another.

Organic participants also spent significantly more total time with the browser tab inactive, which points to more real-world interruptions during the session. Tab switch counts did not differ significantly between groups. Prolific participants likely kept more consistent conditions throughout, given the platform's completion norms. Either way, higher within-person variability among organic participants increases the residual variance in the model, which means wider confidence intervals and less power to detect a treatment effect. So the null result doesn't necessarily mean there's no effect — it may just mean we didn't have enough precision to detect a small one.

0.8.2 Primary Model: Effect of Difficulty Labels on Log Time Engaged

0.8.2.1 Outcome Distribution Raw time engaged is strongly right-skewed, with a long tail from participants who spent disproportionately long on a single puzzle. Log transformation is applied to reduce skew and stabilize variance before modeling.

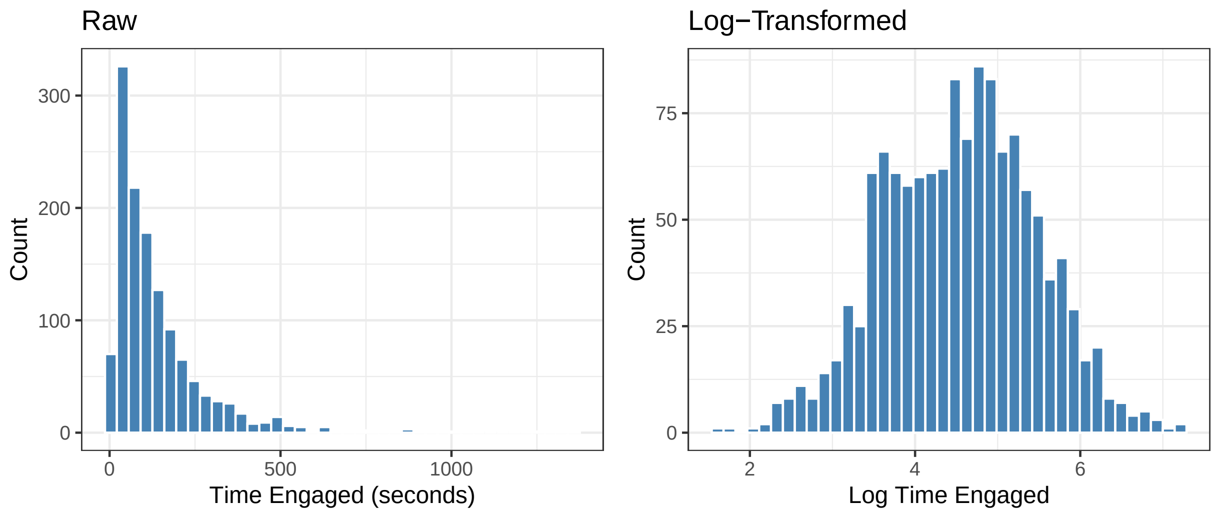

Figure 2: Distribution of time engaged before and after log transformation.

As shown in Figure 2, the log transformation substantially reduces skew and produces an approximately symmetric distribution.

0.8.2.2 Model Specification We estimated the effect of difficulty labels on log time engaged using a linear mixed-effects model with puzzle fixed effects, a participant-level random intercept, and pre-treatment covariates:

$$\log(\text{TimeEngaged}_{it}) = \beta_0 + \beta_1 \text{EasyLabel}_i + \beta_2 \text{HardLabel}_i + \mathbf{X}_i' \boldsymbol{\delta} + \varepsilon_{it}$$

where $\mathbf{X}_i$ includes puzzle skill, recruitment source, age, quantitative exposure, energy level, and device type. Standard errors are clustered at the participant level using CR1 with Satterthwaite degrees of freedom.

Table 3: Effect of Difficulty Labels on Log Time Engaged

	Full	Organic	Prolific
Easy Label	0.003	−0.054	0.069
	(0.076)	(0.106)	(0.105)
	p = 0.965	p = 0.614	p = 0.511
Hard Label	−0.129*	−0.153	−0.122
	(0.077)	(0.111)	(0.114)
	p = 0.095	p = 0.169	p = 0.286

* p < 0.1, ** p < 0.05, *** p < 0.01

Note:
Mixed-effects model (lmer); cluster-robust SEs (CR1) by participant.

Neither label had a statistically significant effect on log time engaged relative to the no-label control in the full sample (Table 3). The easy label estimate is near zero overall ($\hat{\beta}_{\text{easy}}$ = 0.003, SE = 0.076, p = 0.965), but this masks opposing effects across subgroups: organic participants show a negative estimate consistent with H2, while Prolific participants show a slight positive — the two cancel out in the pooled sample. The hard label is consistently negative across both subgroups (organic: -0.153, Prolific: -0.122), and the pooled estimate ($\hat{\beta}_{\text{hard}}$ = -0.129, SE = 0.077, p = 0.097) is marginally significant at the 10% level, corresponding to about 12.1% less time engaged — the opposite direction from H1, which predicted the hard label would increase persistence.

This marginal result should be interpreted cautiously for two reasons. First, across the primary and secondary models we ran multiple comparisons — two treatment coefficients across three samples and two outcomes — and we did not pre-register a correction procedure. Applying a Bonferroni adjustment to the family of tests run here would push the threshold well below the conventional 0.05 level, leaving no result significant. Second, our power analysis targeted 120 participants per arm to achieve 80% power, but we ended up with roughly 85 per arm among completers. At that sample size the simulation suggests power closer to 65%, meaning we likely lack the precision to reliably detect effects of the size we hypothesized, let alone smaller ones.

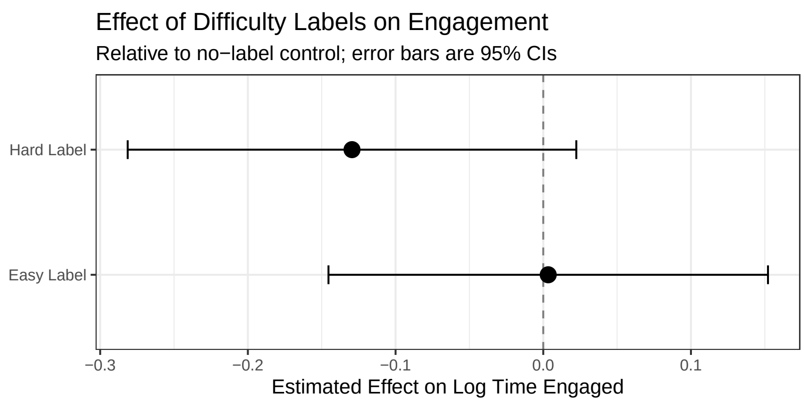

Figure 3: Point estimates and 95% confidence intervals for the effect of difficulty labels on log time engaged relative to the no label control. Error bars are based on cluster-robust standard errors.

As shown in Figure 3, the confidence intervals for both estimates are wide and cross zero, so we can't say much about the true direction of either effect. H3 is also not supported — the hard label point estimate is actually below the easy label estimate, the opposite of what we predicted.

0.8.3 Secondary Model: Effect of Difficulty Labels on Zero-Attempt Skips

0.8.3.1 Outcome Description The secondary outcome is zero-attempt skips — rounds where a participant skipped without entering any response, regardless of how much time or cognitive effort preceded that decision.

Table 4: Zero-attempt skip rate by puzzle and treatment arm.

Puzzle	No Label	Easy Label	Hard Label
Alphabet Wheel	11.2%	6.2%	13.6%
Hybrid Puzzle	0%	0%	0%
Number Sequence	24.7%	18.8%	25.8%
Shape Sequence	1.1%	0%	2.2%
Tile Sequence	22.5%	13.6%	23.6%
Overall	**11.9%**	**7.7%**	**13.1%**

0.8.3.2 Model Specification We model zero-attempt skips using a Linear Probability Model (LPM), estimating the probability of a zero-attempt skip as a linear function of treatment assignment and covariates:

$$\text{ZeroAttSkip}_{it} = \beta_0 + \beta_1 \text{EasyLabel}_i + \beta_2 \text{HardLabel}_i + \mathbf{X}_i' \boldsymbol{\delta} + \varepsilon_{it}$$

where $\mathbf{X}_i$ includes puzzle skill, age, quantitative exposure, energy level, and device type. The full-sample model additionally controls for recruitment source. Standard errors are clustered by participant (CR1, Satterthwaite degrees of freedom), and coefficients are interpreted as percentage-point differences in zero-attempt skip probability relative to the no-label control.

Zero-attempt skips are heavily concentrated in two puzzles. As shown in Table 4, Number Sequence and Tile Sequence account for the vast majority of cold skips, with overall rates of 23% and 20% respectively, while Shape Sequence and Hybrid Puzzle see virtually none. Alphabet Wheel falls in between at around 10%. The easy label arm consistently shows lower skip rates across puzzles, while hard label is slightly elevated relative to no label — a pattern that holds across puzzle types rather than being driven by any single puzzle.

Table 5: Effect of Difficulty Labels on Zero-Attempt Skips

	Full	Organic	Prolific
	(1)	(2)	(3)
Easy Label	-0.039^{*}	-0.019	-0.055^{**}
	(0.021)	(0.034)	(0.025)
	p = 0.068	p = 0.574	p = 0.033
Hard Label	-0.001	0.002	-0.008
	(0.027)	(0.037)	(0.039)
	p = 0.981	p = 0.951	p = 0.841
Observations	1,292	674	618
R^2	0.114	0.133	0.096
Adjusted R^2	0.106	0.118	0.079

Note: *p<0.1; **p<0.05; ***p<0.01
LPM; cluster-robust SEs (CR1) by participant.

The linear model estimates the effect of difficulty labels on the probability of skipping a puzzle without any answer attempt (Table 5). The hard label shows no meaningful effect no matter how we slice the sample — the coefficient is near zero and non-significant in the full sample ($\hat{\beta}_{\text{hard}}$ = -0.001, p = 0.981), among organic participants, and among Prolific participants.

The easy label effect on zero-attempt skips is driven entirely by Prolific participants. Among Prolific participants, the easy label reduces the zero-attempt skip rate by 5.5 percentage points (p = 0.033), while among organic participants the effect is small and non-significant ($\hat{\beta}_{\text{easy}}$ = -0.019, p = 0.573). When the two groups are pooled and recruitment source is controlled for, the organic null attenuates the Prolific signal, yielding a marginally significant estimate of 3.9 percentage points (p = 0.068) in the full sample.

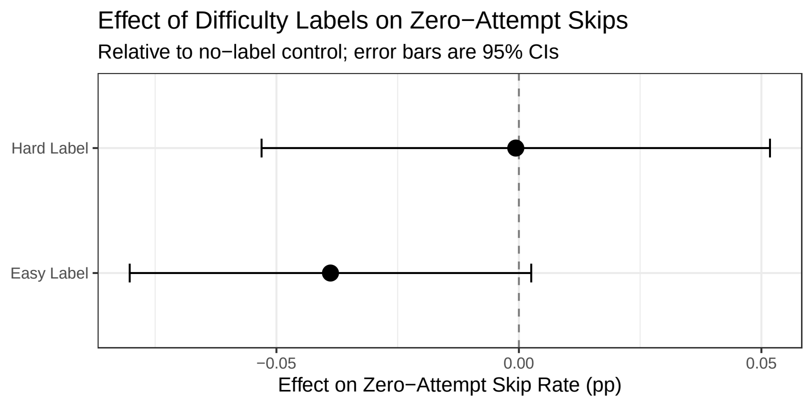

Figure 4: Estimated effect of difficulty labels on zero-attempt skip probability (linear probability model). Coefficients represent percentage-point change relative to the no-label control. Error bars are 95% CIs based on cluster-robust SEs.

As shown in Figure 4, the easy label coefficient is negative and pointing toward fewer cold skips relative to no label. However, the confidence interval is wide and crosses zero in the full sample. The hard label estimate is close to zero with a wide interval in both directions, so there's no real signal there.

The difference in the easy label's effect between Prolific and organic participants likely reflects differences in how each group engages with task framing. Prolific participants are experienced study participants who tend to read instructions carefully and take task cues seriously — when told a puzzle is easy, they internalize that label and may feel a stronger obligation to at least attempt an answer before skipping. Organic participants, recruited informally through personal networks, tend to be more distracted and less attentive to framing; we also found they spent significantly more time with the browser tab inactive, suggesting more real-world interruptions that could override any label effect. In short, the easy label appears to work by making cold-skipping feel less acceptable, but that mechanism requires the participant to be paying attention to the framing in the first place.

0.8.4 Secondary Model: Effect of Difficulty Labels on Time Before Skipping

0.8.4.1 Outcome Description

0.8.4.2 Model Specification Similar to the primary model, we estimated the effect of difficulty labels on log time engaged before skipping. This is a linear mixed-effects model with puzzle fixed effects, a participant-level random intercept, and pre-treatment covariates:

$$\log(\text{TimeEngagedBeforeSkip}_{it}) = \beta_0 + \beta_1 \text{EasyLabel}_i + \beta_2 \text{HardLabel}_i + \mathbf{X}_i' \boldsymbol{\delta} + \varepsilon_{it}$$

Table 6: Effect of Difficulty Labels on Time Before Skipping

	Full	Organic	Prolific
	(1)	(2)	(3)
Easy Label	0.063	0.184	−0.009
	(0.122)	(0.174)	(0.159)
	p = 0.606	p = 0.297	p = 0.955
Hard Label	−0.136	0.021	−0.272*
	(0.108)	(0.147)	(0.156)
	p = 0.210	p = 0.887	p = 0.088
Observations	537	224	313

Note: *p<0.1; **p<0.05; ***p<0.01
Mixed-effects models; CR1 cluster-robust SEs by participant.

0.9 Secondary Model: Effect of Difficulty Labels on Attempt Count

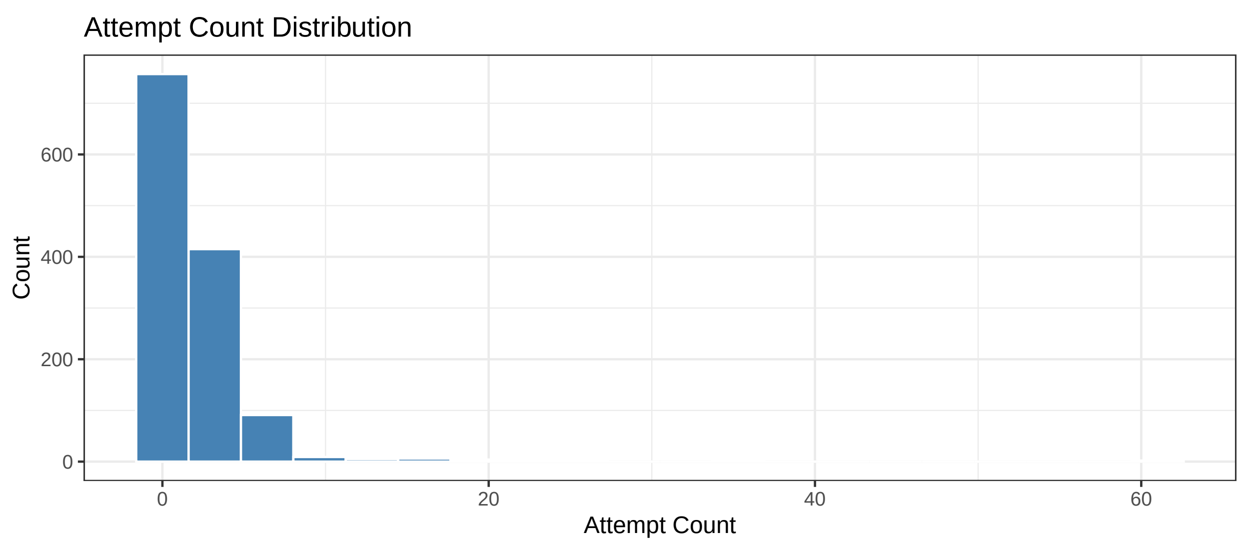

Figure 5: Distribution of Attempt Count Before and After Log Transformation.

0.9.0.1 Outcome Distribution

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    1.00    1.00    2.14    2.00   61.00
```

Attempt count records the number of attempts made on a question before getting the correct answer of skipping. Raw attempt count is strongly right-skewed, with a sparse tail from the few participants who made many attempts on a single puzzle. This is a count variable that only takes integer values. This suggests that a traditional linear model is not appropriate.

0.9.0.2 Model Specification Since our outcome variable, attempt count, is an integer count, traditional linear models are not appropriate. The variance is much larger of our than the mean, (12.0647895»2.1400929), and there is no zero inflation, (proportion of zeroes is 0.1099071). Given these facts, negative Binomial regression is most suited for this task. It is not mathematically possible to use a negative binomial regression model and with cluster robust SEs and a participant level random baseline. If we include userId as a factor, the model estimates a separate intercept for each user (factor(userId)) plus a treatment effect that is constant within each user we elected to use participant as a factor so we can still use clustered robust SE Since treatment is perfectly determined by user (each user has only one treatment), the model is effectively trying to estimate one dummy per user plus a dummy for treatment that is just a relabeling of users which leads to perfect/near-perfect collinearity. This means the treatment effect will not be correctly identified. This leads us to drop the userId covariate entirely. We estimated the effect of difficulty labels on log time engaged using a negative binomial model with puzzle fixed effects and pre-treatment covariates:

$$E[\text{attemptCount}_i] = \exp\Big(\beta_0+\beta_1\,\text{treatment}_i+\beta_2\,\text{puzzleId}_i+\beta_3\,\text{puzzleSkill}_i+\beta_4\,\text{source}_i+\beta_5\,\text{age}_i+\beta_6\,\text{quantitativeExposure}_i+\beta_7$$

Table 7: Effect of Difficulty Labels on Attempt Count

	Full	Organic	Prolific
Easy Label	0.044	−0.089	0.162
	(0.170)	(0.297)	(0.175)
	p = 0.795	p = 0.765	p = 0.354
Hard Label	−0.179	−0.205	−0.025
	(0.128)	(0.206)	(0.125)
	p = 0.161	p = 0.320	p = 0.843
Puzzle 2	0.694***	0.607***	0.776***
	(0.101)	(0.148)	(0.129)
	p = <0.001	p = <0.001	p = <0.001
Puzzle 3	0.548***	0.435**	0.628***
	(0.115)	(0.203)	(0.107)
	p = <0.001	p = 0.032	p = <0.001
Puzzle 4	0.178	0.191	0.181**
	(0.109)	(0.204)	(0.086)
	p = 0.102	p = 0.351	p = 0.035
Puzzle 5	0.284**	0.339	0.235**
	(0.136)	(0.236)	(0.111)
	p = 0.036	p = 0.151	p = 0.035

* p < 0.1, ** p < 0.05, *** p < 0.01

Note:
negative binomial model; cluster-robust SEs (CR1) by participant.

We estimated Negative Binomial regression models to evaluate whether assigning difficulty labels (“Easy” or “Hard”) influences the number of attempts participants make on each puzzle. Models were fit separately for the full sample, organically recruited participants, and participants recruited through Prolific. All models include controls for puzzle identity, puzzle skill rating, source, age, quantitative exposure, energy level, and device type. Standard errors are clustered by participant to account for repeated observations.

Difficulty Label Effects: Across all three samples, neither the Easy Label nor the Hard Label produced statistically detectable changes in attempt count relative to the Unlabeled baseline condition. Coefficients for both labels are small in magnitude and accompanied by large standard errors, indicating substantial uncertainty around the estimated effects. Assigning puzzles an Easy or Hard difficulty label did not meaningfully alter how many attempts participants made.

Puzzle Effects: Puzzle identity shows consistent and substantial associations with attempt count. Relative to the baseline puzzle(puzzle 1), Puzzle 2 and Puzzle 3 exhibit large positive coefficients across all samples, indicating significantly higher expected attempt counts. These effects are statistically significant at the 1 percent level in every group. Puzzle 4 and Puzzle 5 also show positive associations, though their significance varies by sample. All these effects positive with respect the puzzle1, the baseline, meaning they had more attempts than puzzle1. The interpretation of the significant effects are as follows. Puzzle 2 had 2.002% more attempts than the baseline puzzle in the full pool of participants, 1.835% more attempts than the baseline puzzle in the organic participant group, and 2.173% more attempts than the baseline puzzle in the prolific participant group. Puzzle 3 had 1.73% more attempts than the baseline puzzle in the full pool of participants, 1.545% more attempts than the baseline puzzle in the organic participant group, and 1.874% more attempts than the baseline puzzle in the prolific participant group. Puzzle 4 had 1.198% more attempts than the baseline puzzle in the prolific participant group. Puzzle 5 had 1.328% more attempts than the baseline puzzle in the full pool of participants, and 1.265% more attempts than the baseline puzzle in the prolific participant group.

All in all, difficulty labels did not influence how many attempts participants made, regardless of recruitment source. However, puzzle identity consistently predicted attempt count, with some puzzles eliciting substantially more attempts than others. These findings indicate that puzzle-specific features, rather than externally assigned difficulty labels, shape participant behavior in this task.

0.10 Appendix

0.10.1 A. Residual Diagnostics

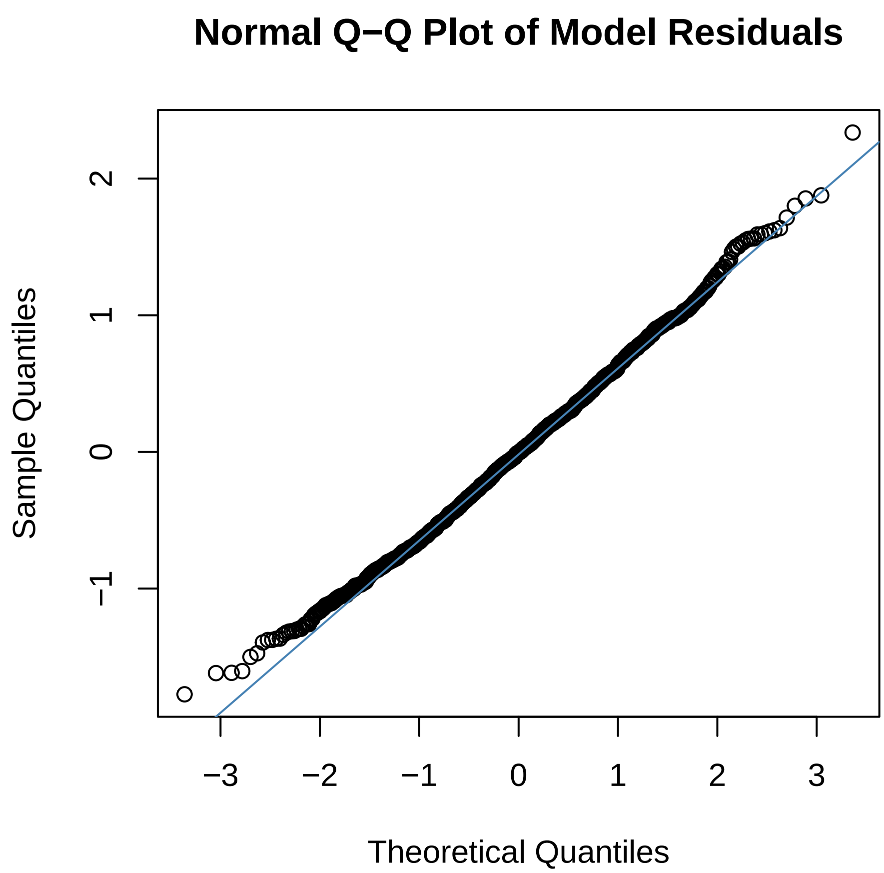

Figure 6: Normal Q-Q plot of residuals from the primary model (log time engaged ~ difficulty label + puzzle fixed effects + covariates + participant random intercept). Residuals track the reference line closely across the central range, with minor deviation in the lower tail, consistent with a light left tail after log transformation.

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